Protecting Smart Grids with Machine Learning

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Introduction

This study aims to create efficient strategies for detecting cyberattacks in smart grids. By applying dimensionality reduction techniques to a dataset of contemporary cyberattacks, the research improves anomaly detection model accuracy by simplifying computations, models, and reducing overfitting. This approach enhances pattern recognition by emphasizing relevant features and eliminating noise, potentially leading to more effective and economical smart grid security solutions.

Methodology

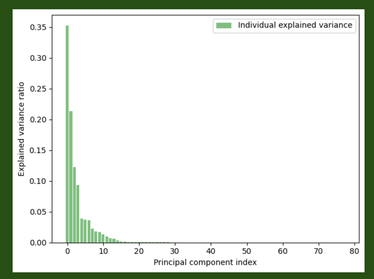
Datasets:
The eight datasets examined comprised widely-known cyber attacks, including Brute Force FTP, Brute Force SSH, DoS, Heartbleed. In total, the datasets included more than 2.5 million benign instances and over 500,000 instances of attacks.

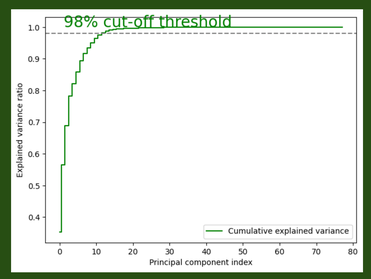
Preprocessing:
To prepare our data for training and testing, we consolidated all the datasets, standardized the data, and selected 100,000 rows of benign instances for training our anomaly detection models. Additionally, we sampled 50,000 rows containing both benign and attack instances for testing purposes.

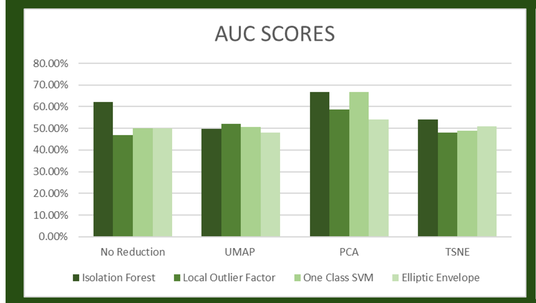
Feature Extraction:
We employed Principal Components Analysis (PCA) to identify the cumulative variance in our data, discovering that only 13 components (i.e., features) were necessary to represent 98% of the data cumulative explained variance. We configured our feature extraction methods (PCA, t-SNE, UMAP) to extract this specific number of features.

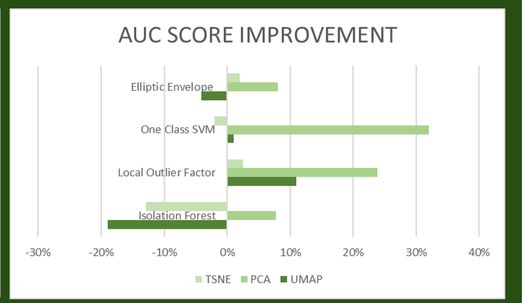
Training and Testing:
The study utilized two semi-supervised anomaly detection algorithms [1]: Isolation Forest and One-Class Support Vector Machine (SVM). Isolation Forest identifies anomalies based on their rarity and distinctiveness, while One-Class SVM detects deviations from 'normal' instances. Additionally, unsupervised methods including Clustering-based Local Outlier Factor, Local Outlier Factor, and Elliptic Envelope were explored. All algorithms were trained using both the complete feature set and a reduced feature set for comparison. The training dataset exclusively contained benign instances, helping the models in understanding normal data patterns. For testing, a dataset comprising both normal and abnormal (i.e., attack) instances was used.

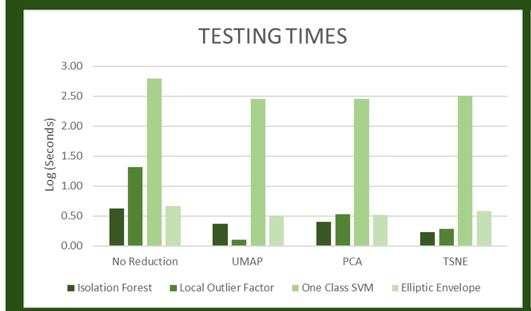
Results

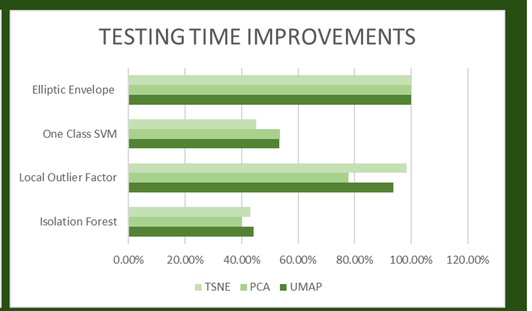


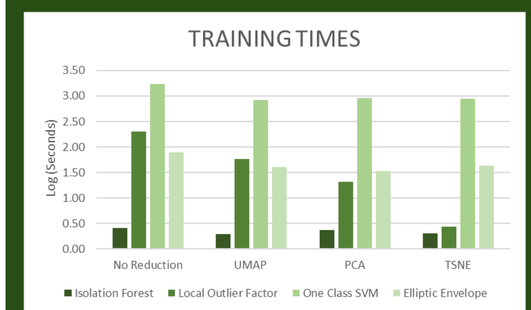


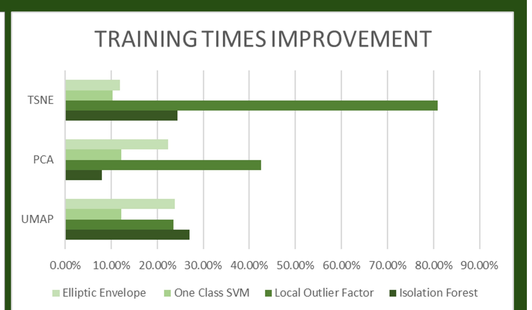












Conclusions

Our research explored various feature extraction methods in anomaly detection algorithms, aiming to enhance efficiency and accuracy. Notably, PCA consistently outperformed other techniques, yielding superior results. Surprisingly, feature extraction did not consistently improve algorithm efficiency, as indicated by the area under the curve (AUC) scores.

Our findings highlight the importance of reducing the number of components. While it didn't guarantee AUC score improvement, it did enhance overall AUC scores in the implemented anomaly detection algorithms. This suggests that the initial data contained unnecessary noise, which, when reduced, led to more accurate anomaly detection.

Additionally, this reduction significantly improved training times, showcasing the practical advantages of optimizing data dimensionality. These insights not only refine our understanding of feature extraction methods but also contribute valuable strategies for enhancing the effectiveness and efficiency of anomaly detection algorithms applied to smart grids.

Future Work

In our future research, we plan to explore different methods to simplify data, such as Variational Autoencoders (VAE). VAEs are great at condensing complex data into simpler forms. We'll compare them with the methods we discussed earlier. Choosing the right method depends on the specific task and data. We want to see if using VAEs can make our anomaly detection models even better.

Acknowledgements

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References

[1] Qi, R.; Rasband, C.; Zheng, J.; Longoria, R. Detecting Cyber Attacks in Smart Grids Using Semi-Supervised Anomaly Detection and Deep Representation Learning. Information 2021, 12, 328. doi: https://doi.org/10.3390/info12080328